

Lucas Yao

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EDUCATION

New York University, New York, NY

MS, Data Science

Expected May 2027

Coursework:

- Introduction to Data Science, Optimization and Computational Linear Algebra, Learning from Small Data, Data Machine Learning, Big Data, Advanced Python for Data Science, Visualization for ML, Natural Language Processing

John Jay College of Criminal Justice, City University of New York (CUNY), New York, NY

BS, Computer Science & Information Security, Minor in Mathematics |

May 2025

Coursework:

- Object-Oriented Programming (C++), Data Structures & Algorithms, Probability & Statistics

PROJECTS

Goodreads RecSys & Market Segmentation | Big Data Capstone

Jan 2026 – May 2026

- Processed a massive interaction dataset of 876K users and 223M rows by migrating raw CSV files to optimized Parquet formats on an HDFS Dataproc cluster to scale Spark computation.
- Engineered a distributed MinHash-based LSH pipeline to execute large-scale similarity searches, successfully identifying and ranking the top 100 most similar pairs of books.
- Optimized Alternating Least Squares (ALS) models utilizing Spark MLlib ranking metrics, tuning explicit-feedback ranks, and synthesizing implicit preference features (is_read, is_reviewed).

Predicting Building Energy Consumption | Controlled ML Comparison

Jan 2026 – May 2026

- Co-authored a benchmark study evaluating 10 machine learning methods across 4 model families on the 20.2M row ASHRAE dataset while keeping features and splits constant.
- Conducted comprehensive feature ablation proving that removing lag and rolling statistics degraded tree ensemble performance by 400-615% compared to only 8-12% for linear models.
- Engineered a global LSTM with a 168-hour lookback using a streaming pipeline to bound memory across 970 meters, outperforming all linear baselines and ranking third overall.

MathLABS | [StreamlitApp](#)

Sep 2025 – Dec 2025

- Collaborated in a team of 4 to construct MathLABS, a small-data benchmark comprising over 400 visual discrete mathematics problems (combining textbook-derived and AI-generated items) for stress-testing Multimodal Large Language Models (MLLMs).
- Engineered the data pipeline and unified JSON schema for all questions, storing question instances, metadata, and associated figures in MongoDB Atlas + HuggingFace Datasets, which support fine-grained analysis of MLLM performance.
- Engineered a hybrid evaluation framework (MathLabs Evaluator) utilizing the Gemini 2.5 Flash model for ground truth validation and an OpenAI-compatible API structure (OpenRouter and Hugging Face Routers) to benchmark a total of nine diverse Vision-Language Models (VLMs) on Math MCQs.

EXPERIENCE

College Assistant | BMCC | New York, NY

Sep 2023 - Dec 2023

- Assisted Computer Science 101 professor with lectures and provided one-on-one tutoring for **20** students.

BMCC Technology Learning Community Intern | BMCC | New York, NY

May 2023 – Jul 2023

- Implemented key features such as a curriculum search function, live chat support, and data analysis visualizations, providing comprehensive resources for academic and career guidance and aiding in informed decision-making for **~1,500** students in the CIS department.

LEADERSHIP / ACTIVITIES

CodePath - Apprenticeship - 102 Web development, 102 Technical Interview Prep

Jun 2025 – Aug 2025

John Jay Computer Science Society Club - Web Developer, CSS [Website](#)

Sep 2024 – May 2025

CUNY Tech Prep Data Science Fellow

Jul 2024 – May 2025

CUNY Peer Leadership Program

Sep 2022 – May 2023

CERTIFICATIONS

- Google Data Analytics Certificate, Coursera, June 2024 - [Link](#)
- Databases for Data Scientists Specialization, Coursera, July 2024 - [Link](#)